

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Civil / Mechanical) (Credit System) Degree Examinations

April 2023

21CV301 / 21ME301 – ENGINEERING MATHEMATICS - III

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

Marks BT* CO* PO*

1. a) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = t - 5$ where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\hat{i} - 2\hat{j} + 2\hat{k}$.
6 L*2 1 2
- b) Prove that the cylindrical polar coordinate system is an orthogonal curvilinear coordinate system.
7 L3 1 1
- c) Show that the vector field $\vec{F} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is conservative. Find its scalar potential.
7 L3 2 2
2. a) Find the directional derivative of $\phi = 3x^2 + 2y - 3z$ at $(1, 1, 1)$ in the direction of $2\hat{i} + 2\hat{j} - \hat{k}$.
6 L2 1 2
- b) State and prove Green's theorem in the plane.
7 L3 2 1
- c) Find the total work done in moving a particle in a force field given by $\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$.
7 L2 2 2
3. a) If $\phi = 2x^3y^2z^4$ and $\vec{F} = \nabla\phi$ then find i) $\text{curl}(\vec{F})$ ii) $\text{div}(\text{grad } \phi)$.
6 L1 1 2
- b) i) Define solenoidal and irrotational vector fields.
ii) Find the constants a, b so that $\vec{F} = (2xy + 3yz)\hat{i} + (x^2 + axz - 4z^2)\hat{j} + (3xy + 2byz)\hat{k}$ is irrotational.
7 L2 1 2
- c) Using Stoke's theorem, evaluate $\int_C (x + y)dx + (2x - z)dy + (y + z)dz$ where C is the boundary of the triangle with vertices $(2, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 6)$.
7 L3 2 2

Unit – II

4. a) Determine the analytic function $f(z)$ whose real part is $e^{-x}(x \sin y - y \cos y)$.
6 L2 3 2
- b) If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then prove that at any point of D , $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.
7 L3 3 1
- c) Using Cauchy's residue theorem, evaluate $\int_C \frac{z^3}{(z-1)^2(z-2)} dz$ where C is the circle $|z| = 2.5$.
7 L2 4 2

5. a) Find the poles and residues at each pole for the function

$$f(z) = \frac{z}{(z+1)(z-2)^2}$$

- b) State and prove Cauchy's integral theorem.

- c) Find the bilinear transformation which maps the points $z=1, i, -1$ onto $w=0, 1, \infty$.

6 L2 4 2
7 L2 4 1

7 L2 3 2

6. a) Show that $u = x^2 - y^2 - y$ is harmonic. Also find v such that $f(z) = u + iv$ is analytic.

6 L2 3 2

- b) Evaluate $\int_C \frac{z^2 - z + 1}{z - 1} dz$ where C is the circle i) $|z| = 2$ ii) $|z| = \frac{1}{2}$.

7 L2 4 2

- c) Find the Laurent's expansion of $\frac{1}{(z-1)(z+3)}$ in the region $1 < |z| < 3$.

7 L2 4 2

Unit – III

7. a) Find the half range Fourier cosine series of $f(x) = \pi - x$, in $0 < x < \pi$.

6 L2 5 2

- b) Obtain the Fourier series expansion of $f(x) = 4 - x^2$ in $(-2, 2)$ and $f(x+4) = f(x)$.

7 L2 5 2

- c) Obtain the Fourier series for the following data by considering the terms up to the first harmonic.

x°	0	60	120	180	240	300
y	9	18	24	28	26	20

7 L2 5 2

8. a) Find the Fourier cosine transform of $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$

6 L2 5 2

- b) State and prove convolution theorem of Fourier transforms.

7 L3 5 1

- c) Find the Fourier transform of $e^{-a|x|}, x \geq 0, a > 0$.

7 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
